

the market. The latest entrant in the Raspberry Pi family is a new variant of the Pi Zero model that was launched at US\$ 5 in November 2015.

Raspberry Pi Zero W widens the use cases of its previous model by bringing wireless connectivity through a built-in Cypress CYW43438 chip. First featured on the Pi 3 Model B, the chip provides 802.11n wireless LAN and Bluetooth 4.0.

"We imagine you will find all sorts of uses for Zero W. It makes a better general-purpose computer because you are less likely to need a hub; if you are using Bluetooth peripherals, you might end up with nothing at all plugged into the USB port," Upton, who heads the non-profit Raspberry Pi Foundation, wrote in a blog post.

Excluding the wireless connectivity support, the newly announced Zero W and the previous Zero have no differences. Both the single-board computers are 3cm (1.2 inch) wide and have a height of 6.6cm (2.6 inches). Also, they are powered by the same BCM2835 chipset that has a processing core at a clock speed of 1GHz.

The Raspberry Pi Zero family has 512MB of RAM and a microSD card slot. There is a full-size GPIO header with 40 pins to let you easily begin with your experiments. Additionally, the board has mini-HDMI ports, USB OTG (On-the-Go) ports, microUSB power socket and a CSI camera connector.

Alongside advancing the Pi Zero line-up, the Raspberry Pi Foundation has partnered with Kinneir Dufort and T-Zero to launch an official injection-moulded case. The new case has three interchangeable lids, and is based on the same aesthetics that were developed for the Raspberry Pi 3 case.

Raspberry Pi Zero W is on sale through all Zero distributors at a price of US\$ 10.

Facebook releases Prophet, the open source forecasting tool

Facebook has expanded its presence in the world of open source by releasing an in-house forecasting tool called Prophet. Available in Python and R programming languages, the latest development is designed to provide automated forecasts that are vital for data scientists and analysts.



Forecasting is critical not just for a company like Facebook but even for a startup or a small enterprise. Facebook is using Prophet across many of its applications to produce forecasts for planning and goal setting. Based on an additive model, the open source solution

is mainly used for hourly, daily and weekly observations with at least a few months of history. It also has strong multiple 'human-scale' seasonalities and works best with daily periodicity, with at least one year of historical data.

"With Prophet, you are not stuck with the results of a completely automatic procedure if the forecast is not satisfactory — an analyst with no training in time series methods can improve or tweak forecasts using a variety of easily interpretable parameters," Facebook engineers Sean J. Taylor and Ben Letham wrote in a detailed note.

Facebook's core data science team is maintaining the code of Prophet.

HPE and Red Hat partner to build open source NFV solutions

Hewlett-Packard Enterprise (HPE) and Red Hat are working together to offer network functions virtualisation (NFV) deployment services for communication platform providers (CSPs). The planned infrastructure is designed to be completely open source.

NFV solutions by HPE will incorporate OpenStack and Ceph storage by Red Hat, whereas Red Hat is planning to productise HPE NFV System 1.4. This whole

Microsoft's Amazon Echo competitor features Cortana on Linux

While Cortana is yet to debut on Linux, Microsoft is reportedly building an Amazon Echo competitor with a mix of Cortana and the open source platform. This smart speaker was officially revealed in December last year, and is currently under development by Samsung-owned Harman Kardon.



The Harman Kardon-made device has passed through Wi-Fi certification that suggests the presence of Linux. The certification shows that the speaker, called Harman Kardon with Cortana, has Linux version 3.8.13. Though the listed kernel is a dated one as we have Linux 4.10 on board, it is enough to let developers play around with the Cortana integration.

Microsoft recently started showing its "love" for Linux and the open source community. Last November, the Redmond company brought Linux to its Windows 10 platform and even joined the Linux Foundation.

However, the Linux power behind the smart Cortana speaker is apparently quite distinct. It will enable Microsoft to rival models such as Amazon Echo and Google Home, and let developers build new experiences for the digital assistant. As Linux is massively spread across major Internet of Things (IoT) developments, this is likely to give Microsoft success in the emerging space.

Going forward, Microsoft is apparently aiming to take on Amazon Alexa and Google Assistant, which can be found on the Echo and Home speakers, respectively.